Lab# 04: Security Operations Investigations

Objective: Learn about security operation investigations.

1. SOC RADAR

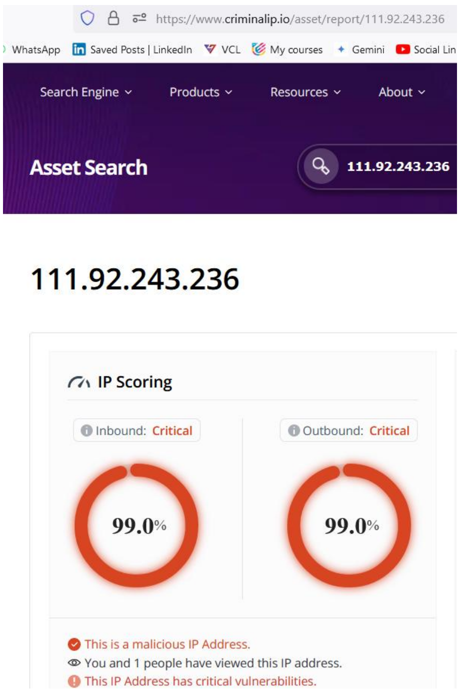
Step 1: Open SOCRadar link from https://socradar.io/labs/soc-tools/ip-reputation to study the various data options to collect artifacts for investigation.

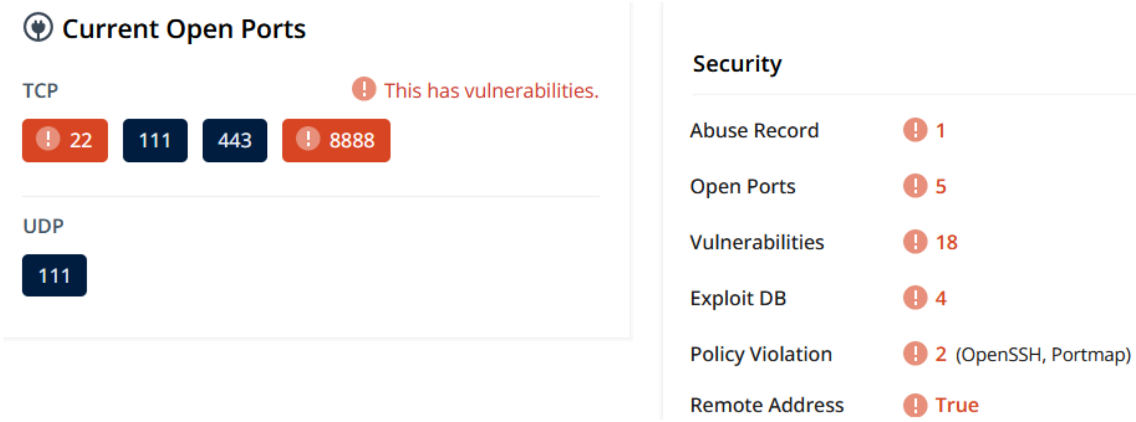
2. Threat View

Step 1: Open link to collect high-confidence IP address from https://threatview.io/Downloads/IP-High-Confidence-Feed.txt

```
#Command and Control Feed Generated by Threatview[.]io Proactive Hunter on 02 June 2024. Domain and IP feed https[:]//threatview
# See any false posstives ? Email us feeds[@]threatview[.]io
#IP,Date of Detection,Host,Protocol,Beacon Config,Comment
1.15.248.225,02 June 2024 07:07 PM UTC,1.15.248.225,https,"1.15.248.225,/__utm.gif",Generated by Threatview[.]io
101.201.54.74,02 June 2024 07:07 PM UTC,101.201.54.74,https,"101.201.54.74,/load",Generated by Threatview[.]io
101.43.32.212,02 June 2024 07:07 PM UTC,101.43.32.212,https,"service-g0t0y6tj-1324325324.cd.tencentapigw.com,/prod/api/debug",Ge
106.54.209.36,02 June 2024 07:07 PM UTC,106.54.209.36,https,"106.54.209.36,/cx",Generated by Threatview[.]io
111.229.187.212,02 June 2024 07:07 PM UTC,111.229.187.212,https,"111.229.187.212,/cm",Generated by Threatview[.]io
111.230.12.238,02 June 2024 07:07 PM UTC,111.230.12.238,https,"111.230.12.238,/wp06/wp-includes/po.php",Generated by Threatview[
111.92.243.236,02 June 2024 07:07 PM UTC,111.92.243.236,https,"111.92.243.236,/claim/servlets-examples/I2I52XQKQQZF",Generated b
113.31.105.33,02 June 2024 07:07 PM UTC,113.31.105.33,https,"service-1bsjckga-1252578700.gz.tencentapigw.com.cn,/api/x",Generate
113.31.106.106,02 June 2024 07:07 PM UTC,113.31.106.106,https,"113.31.106.106,/preserve/Extranet/LFF00FQ6U2H0",Generated by Thre
114.55.133.151,02 June 2024 07:07 PM UTC,114.55.133.151,https,"114.55.133.151,/load",Generated by Threatview[.]io
117.72.8.192,02 June 2024 07:07 PM UTC,117.72.8.192,https,"117.72.8.192,/c/msdownload/update/others/2024/05/9Dv7AyHg1Ag2Kw030_",
118.31.116.9,02 June 2024 07:07 PM UTC,118.31.116.9,https,"118.31.116.9,/jquery-3.3.1.min.js",Generated by Threatview[.]io
119.28.83.149,02 June 2024 07:07 PM UTC,119.28.83.149,https,"python.org,/load",Generated by Threatview[.]io
```

Step 2: Use CriminalIP Portal https://www.criminalip.io from to study these C2 Server IP addresses.





3. Anaylse Emails

Email Headers are metadata that provides information about an email message. They are typically located at the beginning of an email, before the body content. Headers contain various fields that describe the email's origin, destination, and other relevant details:

- **From:** The email address of the sender.
- **To:** The email address of the recipient.
- **Cc:** Carbon Copy recipients.
- **Bcc:** Blind Carbon Copy recipients.
- **Subject:** The subject line of the email.
- **Date:** The date and time the email was sent.
- **Message-ID:** A unique identifier for the email.
- **Received:** A list of servers that handled the email during transit.
- **Return-Path:** The email address used for bouncing undeliverable messages.
- **X-Header Fields:** Custom headers added by email servers or applications (e.g., X-Mailer, X-Spam-Score).

Why are Email Headers Important?

- **Determining Email Origin:** Headers can help identify the source of an email, including the sender's domain and IP address.
- **Investigating Email Fraud:** Headers can be used to trace the path of a fraudulent email, identifying potential spoofers or phishing attempts.
- **Analyzing Email Traffic:** Headers provide insights into email routing, delivery times, and potential issues.
- **Troubleshooting Email Problems:** Headers can help diagnose delivery failures or spam filtering issues.

How to View Email Headers:

- **Most email clients:** Have a built-in option to view headers. Look for options like "View Source," "Show Headers," or "Raw Message."
- **Online tools:** There are many online tools available that allow you to paste an email's content and view its headers.

By understanding email headers, you can gain valuable information about the origin, routing, and content of emails, which is essential for various tasks, including email investigations, security analysis, and troubleshooting.

SPF, DKIM, and DMARC:

SPF, DKIM, and DMARC are three essential authentication protocols that play a crucial role in email investigations. They help verify the sender's identity, preventing spoofing and phishing attacks.

SPF (Sender Policy Framework)

- **Purpose:** Prevents spoofing by specifying which IP addresses are authorized to send emails on behalf of a domain.
- **Significance:**
  - **Identification of Spoofed Emails:** If an email is received from an IP address not listed in the domain's SPF record, it's likely a spoofed email.
  - **Filtering and Blocking:** Email servers can use SPF to filter out spoofed emails and block them from reaching inboxes.

DKIM (DomainKeys Identified Mail)

- **Purpose:** Adds a digital signature to emails, verifying the sender's identity and integrity of the message content.
- **Significance:**
  - **Authenticity Verification:** DKIM ensures that the email originated from the claimed domain and hasn't been tampered with during transit.
  - **Phishing Prevention:** Phishing emails often lack DKIM signatures, making it easier to detect and block them.

DMARC (Domain-based Message Authentication, Reporting, and Conformance)

- **Purpose:** Combines SPF and DKIM to provide a comprehensive framework for email authentication, reporting, and enforcement.
- **Significance:**
  - **Centralized Management:** DMARC allows organizations to manage both SPF and DKIM policies from a single location.
  - **Policy Enforcement:** DMARC can instruct receiving servers to take specific actions, such as quarantining or rejecting emails that fail authentication.
  - **Reporting:** DMARC provides detailed reports on email authentication failures, helping organizations identify and address vulnerabilities.

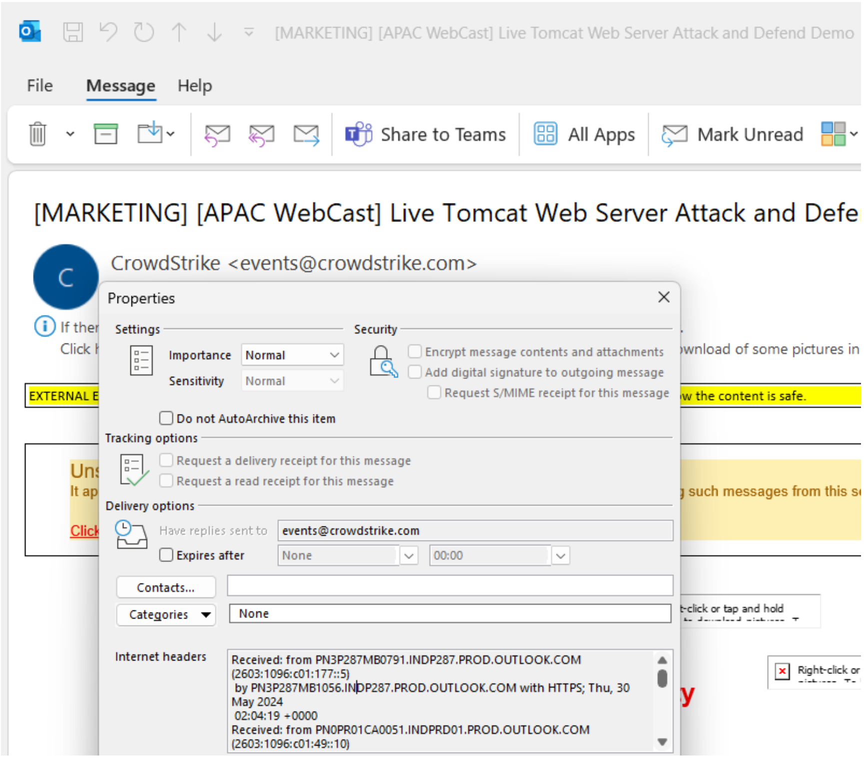
In email investigations:

- **Identifying Spoofed Emails:** SPF and DKIM help identify spoofed emails by verifying the sender's identity and IP address.
- **Detecting Phishing Attempts:** Phishing emails often lack proper authentication, making them easier to detect using SPF, DKIM, and DMARC.
- **Investigating Email Fraud:** These protocols can provide valuable evidence in cases of email fraud by confirming the authenticity of emails and identifying potential spoofers.
- **Improving Email Security:** Implementing SPF, DKIM, and DMARC can significantly enhance email security and protect organizations from various threats.

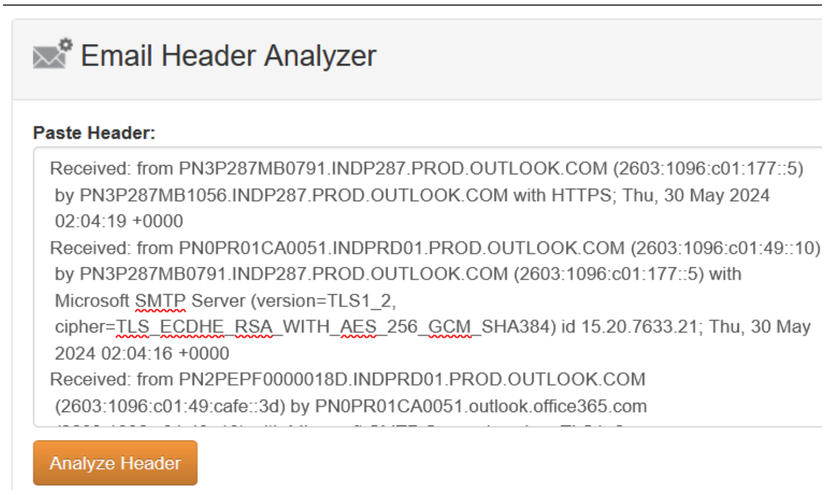
By understanding the significance of these protocols, investigators can effectively analyse email evidence, identify suspicious activity, and take appropriate actions to mitigate risks.

When using Email Apps (like MS Outlook):

Step 1: If using Outlook → Open a spam email → Properties → Internet Headers.



Step 2: Copy all the content displayed in the 'Header information' box and paste into 'Analyze Header' windows of any Email analyzer portal E.g. → https://mxtoolbox.com/EmailHeaders.aspx.

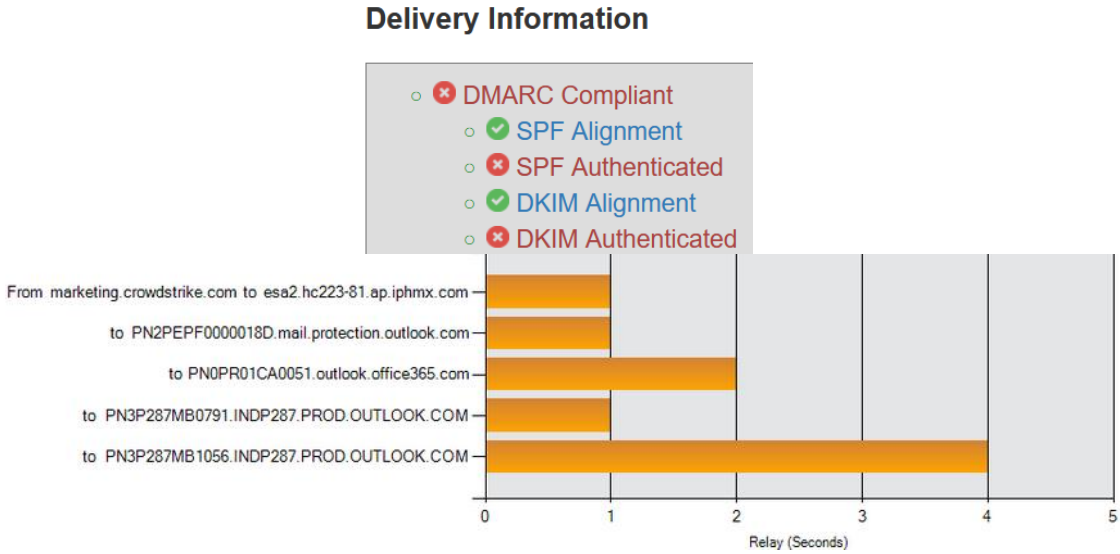


When using Web-based Emails

Step 1: Say you are using Gmail or any Web-based email server, open the email, click on … (three dots) on the right side of the page and select 'Show Original' and click 'Copy to Clipboard'.

Step 2: Open MXToolBox Email-Header from https://mxtoolbox.com/EmailHeaders.aspx and paste the contents into the Email Header Analyzer box and select 'Analyze Header'

Step 3: MXTooBox reports DMARC Compliance and details about the sender and email servers involved. MXToolbox DMARC Compliance refers to the results of DMARC record lookup tool. DMARC (Domain-Based Message Authentication, Reporting, and Conformance) is an email authentication protocol that helps protect against email spoofing and phishing attacks. MXToolbox report indicates the DMARC compliance level based on the policies defined in the DMARC record.



Step 4: SPF and DKIM information are also shared by MXtoolbox when analyzing email headers. SPF and DKIM are two email authentication methods that help prevent spam and phishing attacks.

- SPF (Sender Policy Framework): An SPF record specifies authorized senders for a domain's emails. When an email arrives, the receiving server checks the SPF record of the sender's domain to see if the email originated from a legitimate source. This helps prevent spoofing, where someone sends emails pretending to be from your domain.

SPF and DKIM Information

dmarc:crowdstrike.com Show Solve Email Delivery Problems

```
v=DMARC1; p=reject; fo=1; ri=3600; rua=mailto:crowdstrike@rua.agari.com; ruf=mailto:crowdstrike@ruf.agari.com
```

- DKIM (DomainKeys Identified Mail): DKIM adds a digital signature to emails. This signature allows the receiving server to verify that the email content hasn't been tampered with during transmission. DKIM helps ensure the authenticity of the email sender and the message itself.

spf:marketing.crowdstrike.com:139.138.31.66 Hide

```
v=spf1 include:mktomail.com -all
```

Prefix	Type	Value	PrefixDesc	Description
	v	spf1		The SPF record version
+	include	mktomail.com	Pass	The specified domain is searched for an 'allow'.
-	all		Fail	Always matches. It goes at the end of your record

By analyzing SPF and DKIM information in email headers, MXtoolbox can tell you if the email is likely legitimate or if there might be some red flags. This can be helpful for determining whether an email is spam or phishing attempt.

Lab Activity:

1. Use SOC RADAR to generate reports by collecting at least five artifacts for investigations. List compromised data from Dark Web reports.

2. Use Threat View to gather High-Confidence IP Addresses and use CriminalIP portal to find at least five C2 Servers.

3. Analyse email headers of three different emails – sent from non-Gmail/other email accounts – a. sent by a friend (confirmed), b. sent by an organization, c. sent by spammers. Find the sender's IP and location, sender domain DMAC info and validate if the email is spam or not.